

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B. Tech.

Branch :Automation and Robotics/Robotics/Mechatronics

Engineering

Semester : IV

Subject Code & Name: BTMXC404 Theory of Machines and Mechanisms

Max Marks: 60

Date: 29.01.2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Define inversion of mechanisms? Explain the various inversions of single slider crank chain mechanisms.	1	6
B) What are straight line mechanisms? Describe any one type of straight line mechanism with figure.	2	6
C) Describe transmission angle of mechanism with figure.	2	6
Q.2 Solve Any Two of the following.		12
A) Elaborate the velocity and acceleration analysis using loop closure equations method.	2	6
B) Locate all the instantaneous centers for a four bar mechanisms as shown in Fig 1. The lengths of the various links are: AD = 500 mm; AB = 150 mm; BC = 280 mm; CD = 250 mm. if the AB link rotates at a uniform speed of 300 rpm in the anticlockwise direction, find the angular velocity of the links CD.	2	6

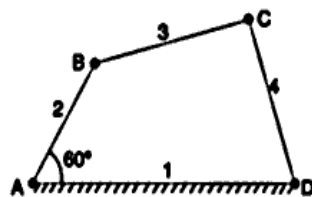


Fig. 1

C) Explain types of instantaneous centers of a mechanisms with figure.	2	6
Q.3 Solve Any Two of the following.		12
A) A shaper mechanism is shown in Fig. 2. The crank PQ is 40 mm long. Length of slotted lever OR is 200 mm. Link RS measures 80 mm and it is used to	2	6

drive the ram. Determine acceleration of slider S for the positions shown when the crank PQ rotates anticlockwise at 40 rpm.

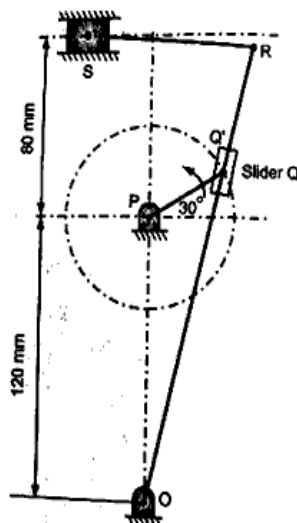


Fig. 2

- | | | |
|---|---|---|
| B) Describe with figure the directional relationship between linear velocity (v) and angular velocity (ω) of Coriolis component of acceleration. | 2 | 6 |
| C) Describe the three positions synthesis graphical method for motion generation. | 2 | 6 |

Q.4 Solve Any Two of the following.

12

- | | | |
|--|---|---|
| A) Derive the expression for law of gearing. | 2 | 6 |
| B) Draw the displacement, velocity and acceleration diagrams for a follower when it moves with simple harmonic motion. | 3 | 6 |
| C) Two mating gears have 20 and 40 involute teeth of module 10 mm and 20° pressure angle. The addendum on each wheel is to be made of such a length that the line of contact on each side of the pitch point has half the maximum possible length. Determine the addendum height for each gear wheel and length of path of contact. | 2 | 6 |

Q. 5 Solve Any Two of the following.

12

- | | | |
|--|---|---|
| A) Derive an expression for the length of belt in an open belt drive. | 2 | 6 |
| B) A single plate clutch, effective on both sides, is required to transmit 25 kW at 3000 rpm. Determine the outer and inner radii of frictional surface if the coefficient of friction is 0.255, the ratio of radii is 1.25 and the maximum pressure is not to exceed 0.1 N/mm^2 . Also determine the axial thrust to be provided by springs. Assume the theory of uniform wear. | 2 | 6 |
| C) Enlist the different types of brakes and explain any one type in detail with figure. | 2 | 6 |

***** End *****

The grid and the borders of the table will be hidden before final printing.